Renty Intelligent P2P Rental Platform

RENTY

Intelligent Peer-to-Peer Rental

RENTY

Egyptians spend up to

10,000 EGP on gear

they use 3 times a year.

There's a smarter way.

96%

open to renting
if they trust the platform

THE PROBLEM

Owning sports gear doesn't make sense.



Too Expensive

A climbing set costs 4,000–10,000 EGP.
Most people can't justify the purchase.



Sits Idle 80% of the Time

Gear collects dust between trips.
Ownership doesn't match usage frequency.



Storage is a Real Burden

Bulky bikes and camping gear don't fit
in typical Cairo apartments.

THE GAP

People want to rent. Nothing safe exists.

Facebook Groups

No protection, cash deals, zero recourse

OLX / Dubizzle

Buy/sell only no rental model

Friends & Network

Limited availability, doesn't scale

50%

already rent or borrow gear informally

The behavior exists.
The infrastructure doesn't.

KEY INSIGHT

We surveyed 254 Egyptians.

96% want to rent.
70% are afraid to.

The barrier isn't demand - it's trust.

Solve trust → unlock the entire market.

INTRODUCING

Renty

Egypt's peer-to-peer sports equipment rental marketplace built around trust.

Rent

Access gear you need without buying it

Lend

Earn from equipment you already own

Trust

Every transaction verified & protected

HOW IT WORKS

Simple for users. Safe by design.

01

Owner Lists

Gear posted with photos & availability

02

Renter Books

Browses, selects, pays securely

03

Escrow Holds

Payment held safely until rental ends

04

Return & Release

Item returned → payment released

No trust required between strangers — the platform holds it.

THE TRUST SYSTEM

These aren't features.
They're why people will actually use it.

Ratings & Reviews

74% demanded this

Every renter and lender rated after each transaction.

Expert Verification

73% feel safer with it

Physical gear inspection at Renty Hubs before rental.

ID Verification

36% require it

Identity confirmed before any transaction is approved.

Deposit Protection

94% expect a deposit

Meaningful deposit held in escrow not a token gesture.

MARKET FOCUS

Narrow on purpose.

This is where density and trust compound fastest.



CATEGORY

Cycling

59.8% demand nearly 2x the next category



SEGMENT

18–24 yr

93% of our validated demand base



LOCATION

Cairo / Giza

80% won't travel more than 15km to pick up

THE VISION

Phase 1

Cycling · Cairo

18–24 · Dense neighborhoods
Verification-first launch

Phase 2

6 Categories · All Egypt

Racket sports, water sports,
camping gear and beyond

Phase 3

MENA Expansion

The infrastructure layer
for sports access in MENA

The Airbnb of sports gear in Egypt.

Starting with
one city,
one category,
one problem.

RENTY

BUSINESS MODEL

How Renty Makes Money

6 revenue streams each validated by real data.

01 PRIMARY

Transaction Commission

82%

would use platform

% cut from every completed rental from renter, owner, or split. Like Airbnb & Turo. High volume makes even small % highly viable.

02 SECONDARY

Insurance Add-on

53.5%

rated it 4–5 / 5

Optional rental insurance per booking. Partnership with insurer (commission model) or in-house margin. Survey avg: 3.7 / 5 value score.

03 SECONDARY

Verification Fee

67.7%

trust expert checks

Per-item inspection fee charged to lister. Critical for high-value gear (bikes, dive equipment). Market explicitly asked for this feature.

04 TERTIARY

Delivery Margin

35.8%

prefer delivery

Margin on platform-arranged logistics via a courier partner. Applied per delivery — turns a user preference into a revenue line.

05 FUTURE

Premium Membership

Recurring

revenue

For power users: lower commission, free insurance, search priority. Best introduced after sufficient transaction volume is established.

06 FUTURE

Featured Listings

Ad model

like e-commerce

Owners pay to appear first in search. Similar to Amazon Sponsored. Best suited after the listings base grows significantly.

Primary streams active at launch · Future streams unlocked as platform scales

254
Respondents surveyed in two weeks

82%
Would rent equipment through Renty (rent-only + both)

96%
Open to using Renty (definitely + maybe)

BOTH — RENT & LEND

Both — rent & lend	45.7%
Rent items only	36.6%
Lend my own items only	9.8%
Neither	7.9%



CATEGORIES MOST LIKELY TO RENT

Cycling (bikes/scooters)	59.8%
Racket sports	31.5%
Scuba diving	24.8%
Camping & hiking	14.2%
Winter sports	11.4%

WHY RENT INSTEAD OF BUY

One-time trip / season	53.1%
Save money	45.7%
Try before buying	31.1%
Too bulky to store	22.4%

⚠ TOP CONCERNS

Item damage during use	70.9%
Poor / unsafe quality	57.5%
Hidden fees	44.5%
Payment security	21.3%

✓ WHAT WOULD BUILD TRUST

Ratings & reviews	74.0%
Secure payment	43.7%
Expert verification	37.4%
ID verification	35.8%

PREFERRED PICKUP / DROP-OFF

Direct meet-up	37.4%
Platform delivery	35.8%
Locker drop-off	21.3%
Designated location	5.5%

THE TAKEAWAY

The same 71% worried about item damage are the 73% who'd trust an expert check before/after rental. Concern and solution were measured independently — and point to the same fix: verification + ratings.

Platform Features

Platform Features

Secure Rentals

Escrow payments
+
deposit protection

Expert Verification

Physical inspection at Renty Hubs

Smart Recommendations

ML-powered personalized suggestions

Real-time Chat

SignalR instant messaging

Fraud Detection

Behavioral anomaly scoring

Gap Analytics

Supply-demand intelligence

Multi-platform

Angular Web + Flutter Mobile

Admin Dashboard

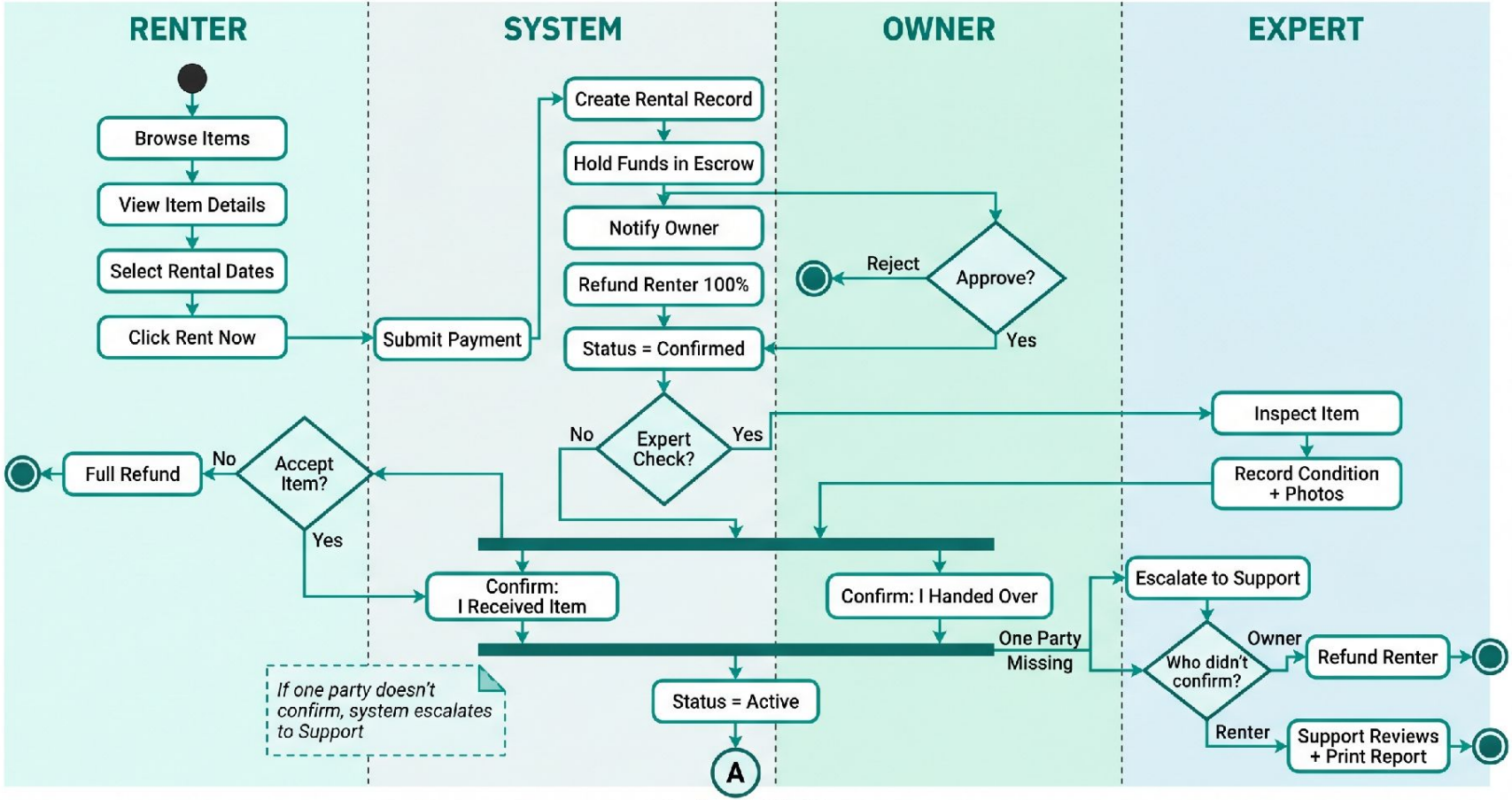
KPIs, user management, moderation

Dispute Resolution

3-tier escalation system

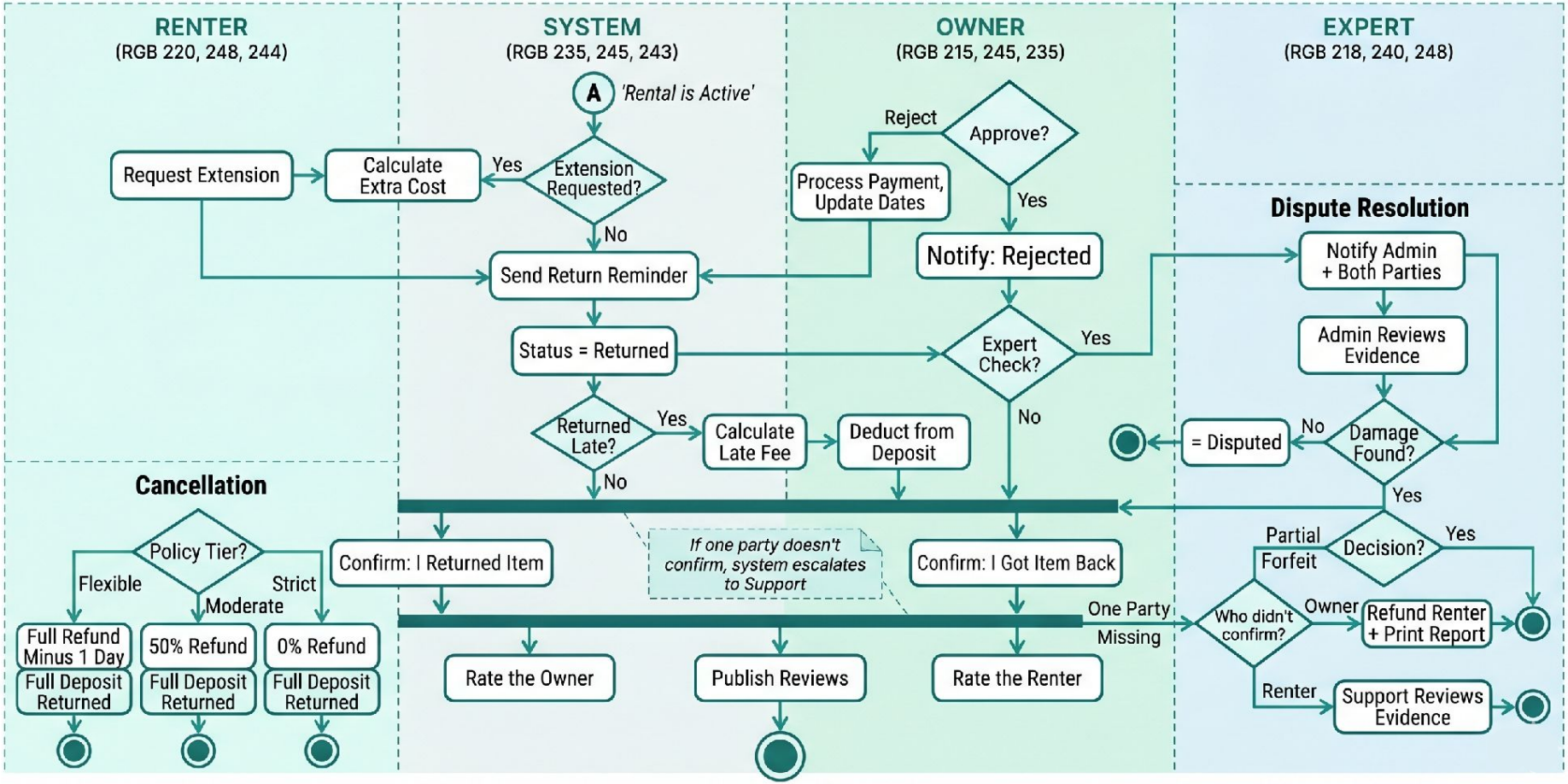
How it works

How it works Rental Flow — Discovery to Handover



Continued in Slide 25b

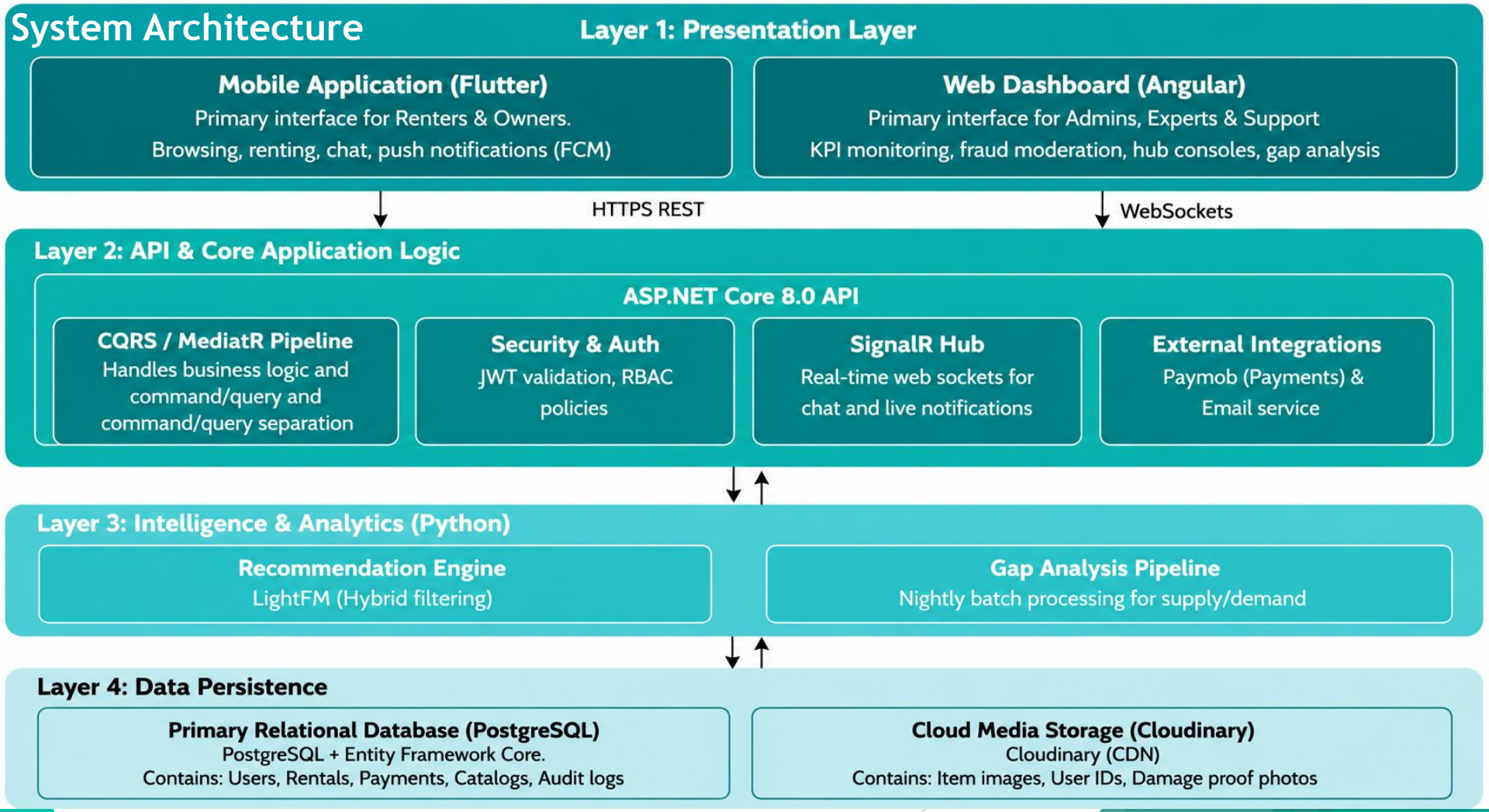
How it works 'Rental Flow — Active Rental to Completion'



Continued in Slide 25b

System Architecture

System Architecture



Tech Stack

Tech Stack

ML / Analytics

- Python 3.11
- LightFM 1.17 (RS)
- scikit-learn 1.4 (Fraud)
- pandas 2.2 (Gap Analysis)

Infrastructure

- PostgreSQL 16
- Cloudinary (CDN)
- Paymob (Payments)
- JWT + RBAC (Auth)

Frontend

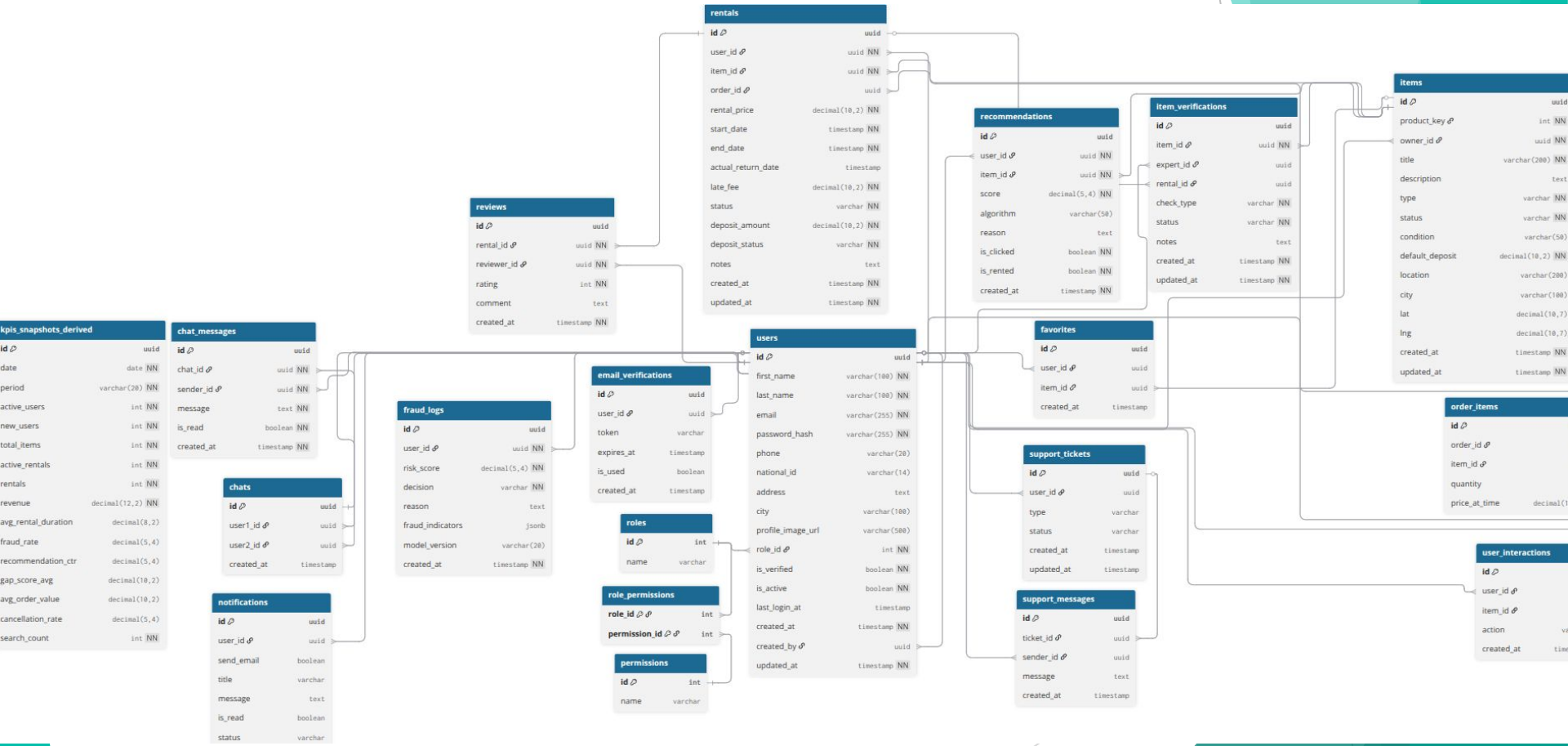
- Angular 19 (Web)
- Flutter 3.x (Mobile)
- Dart (Mobile language)
- Chart.js (Visualization)

Backend

- ASP.NET Core 8.0
- Entity Framework Core 8.0
- MediatR 12 (CQRS)
- FluentValidation 11
- SignalR (WebSockets)
- BCrypt.Net (Security)

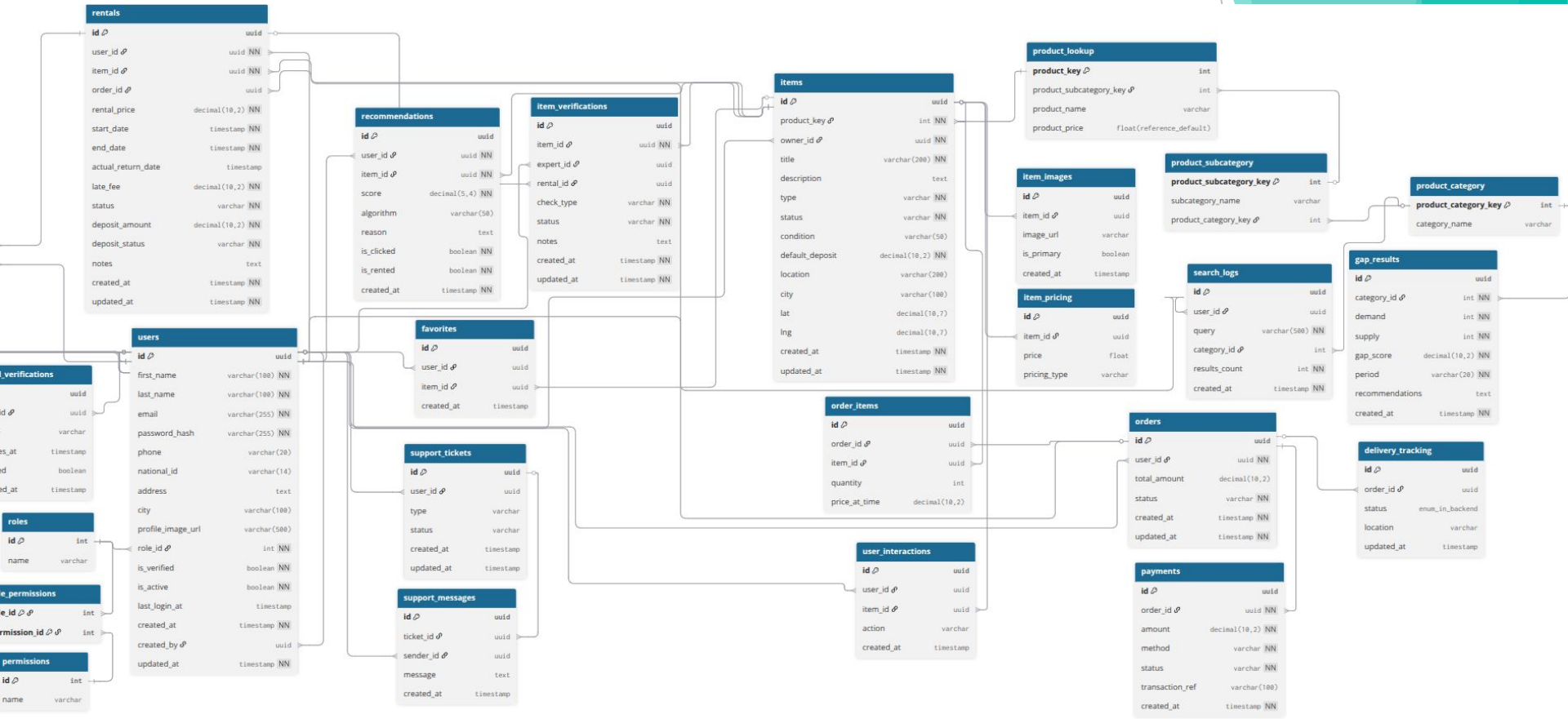
Database Design

Database Design



27 tables | 150+ columns | Full referential integrity

Database Design



27 tables | 150+ columns | Full referential integrity

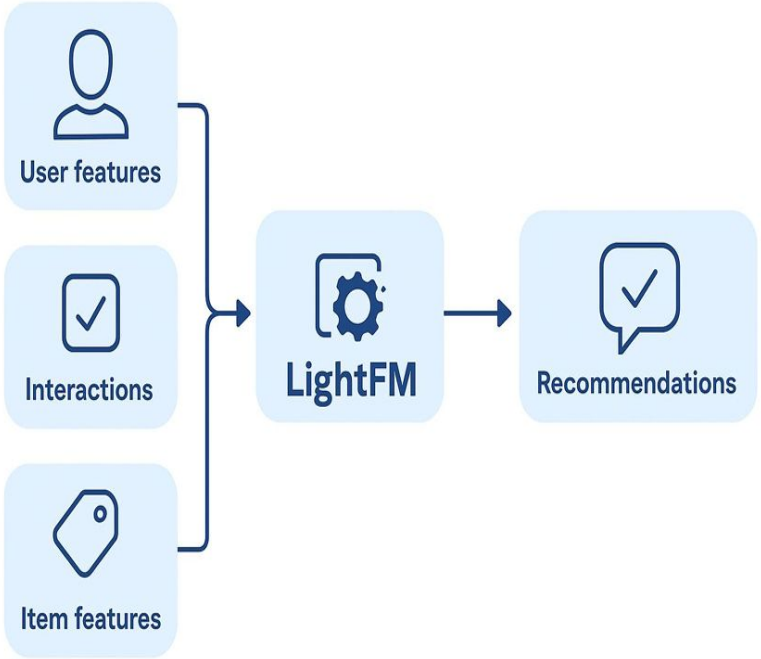
Recommendation System

Why Renty Needs a Recommender System

- Recommend relevant rental items to users based on their profiles and behavior
- Increase user satisfaction and engagement through personalization
- Improve item utilization and revenue for the platform owners
- Reduce cognitive load and decision-making time for users

LightFM

LightFM is a hybrid matrix factorization model that learns latent representations for users and items while integrating user features, item features, and interactions. It supports implicit feedback such as rentals, purchases, and clicks, making it ideal for e-commerce scenarios without explicit ratings.



Results and Performance

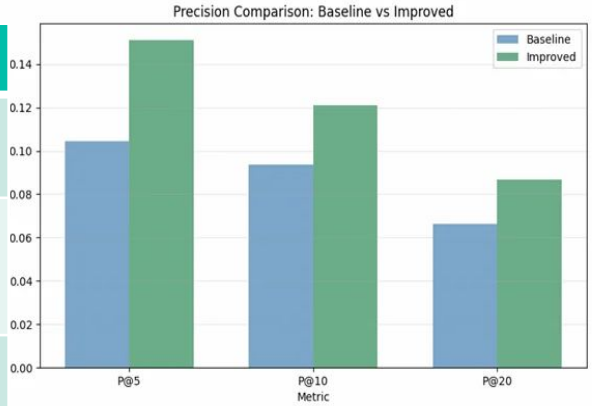
Final Model Performance

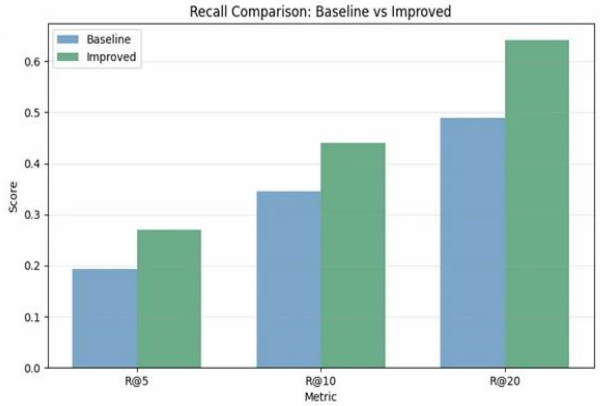
Metric	Baseline	Hybrid	Improvement
Test AUC	**0.7497**	**0.8421**	**+12.3%**
Precision@10	0.0938	0.1208	+28.8%
Overfitting Gap	0.2501	0.0479	-80.8%

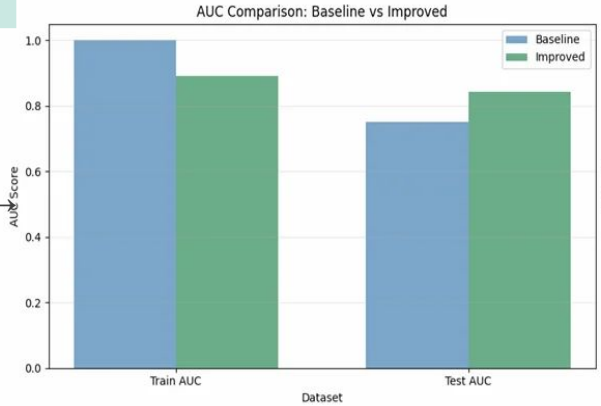
Key Observations:

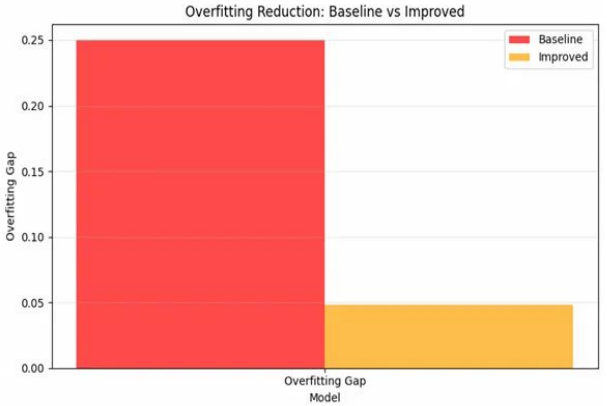
- Precision and Recall consistently higher across all K values
- AUC improved by 12.3% (0.7497 → 0.8421)
- Overfitting gap reduced by 80.8% Model generalizes well to unseen interactions

Performance Visualization









Business Impact

- **Increased Conversion:** Users 29% more likely to purchase recommended items
- **Enhanced Discovery:** 64.2% recall means users discover most relevant products
- **Better User Experience:** Personalized recommendations increase satisfaction
- **Reduced Bounce Rate:** Relevant suggestions keep users engaged
- **Improved Cross-Selling:** Model identifies complementary products

Streamlit app

Notebook

Mobile Application

Why Flutter

Why Choose Flutter?

- Cross-Platform Efficiency: **Single** codebase compiles natively for both Android & iOS, optimizing developer resources.
- Native-Grade Performance: Compiled to machine code via Dart (AOT), providing high responsiveness and smooth transitions (60 FPS).
- Stateful Hot Reload: Significantly accelerated UI prototyping, live bug fixes, and third-party service integration.

Key Integration Stack

- GetX State Management: Implements lightweight, reactive state propagation and clean context-free routing.
- Native Device Capabilities: Smooth hooks into Geolocator (proximity search), Image Picker (item photos), and SharedPreferences (local sessions).
- Rich UI Elements: Out-of-the-box Material Widgets, custom Lottie animations, and localized language handling.

Platform Ecosystem & User Roles

Mobile Client: End-User Focused

- Tailored Roles: Developed exclusively for Renters and Lenders. Administration and operational dashboards are entirely excluded to maximize security and keep the application lightweight.
- On-The-Go Operations: Specialized in high-frequency mobile tasks: browse catalog, message owners, make secure bookings, and check out.
- Seamless User Experience: Simplifies onboarding and transactions with localized interfaces (Arabic & English support).

Web Client & UI Alignment

- Back-Office Operations: Complex analytics, admin panels, dispute resolution, and category configurations are delegated to the Web portal.
- Visual Design Alignment: Mobile UI shares identical color semantics, responsive structures, and typography styles with the Web client to build brand consistency.
- Real-Time Synchronicity: User data and item listings are fully synchronized across Web and Mobile.

Mobile App Feature Matrix

Core Rental Flow

- Marketplace & Search: Explore product cards, keyword search with dynamic categories, and browse hybrid recommendation models.
- Product Management: Add, edit, or remove rental items with title, pricing schemes, deposits, and descriptive tags.
- Transaction Hub: Complete checkouts, track rental lifecycles (Pending, Active, Returned), and manage invoices.

Interactive Utilities

- Peer-to-Peer Chat: Real-time negotiation chat between Renters and Lenders with WebSockets/SignalR integration.
- Trust & Support: Integrated support ticket creation, expert verification histories, and transaction point wallets.
- Accessibility Features: In-app Light and Dark theme toggling and full Arabic/English localization.

Deployment Status & Challenges

Technical Roadblocks

- Late Backend APIs: Delayed delivery of final backend API endpoints restricted full end-to-end integration testing.
- Agile Adaptability: Handled by writing robust mock providers and local storage fallbacks to ensure client stability.
- UI Completeness: Reached a 100% complete and operational frontend layout, verifying all navigation, styling, and static components.

Google Play Store Publishing:

- Production Upload: The app was successfully compiled and uploaded to the Google Play Console in release mode.
- Closed Testing Success: Fully passed Google's strict requirement of 14-day closed testing with 20 active external testers.
- Review State: Testing concluded successfully; currently in the final file revision and review phase awaiting official store release.

Mobile App Demonstration

One Minute Core Flow Walkthrough

Click here to watch the Mobile App Screen Record Demo Video

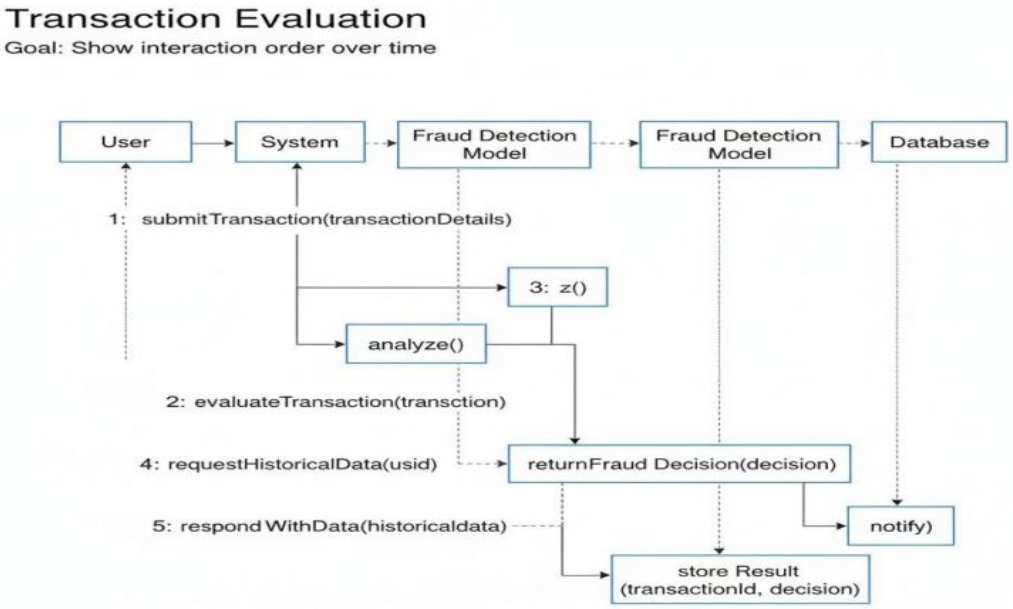
Play Store

Email: rentyprof@gmail.com
Pass: Renty-123



Fraud Detection

Fraud Detection



16 behavioral features — Temporal, Velocity, Financial, Geographic16 behavioral features — Temporal, Velocity, Financial, Geographic

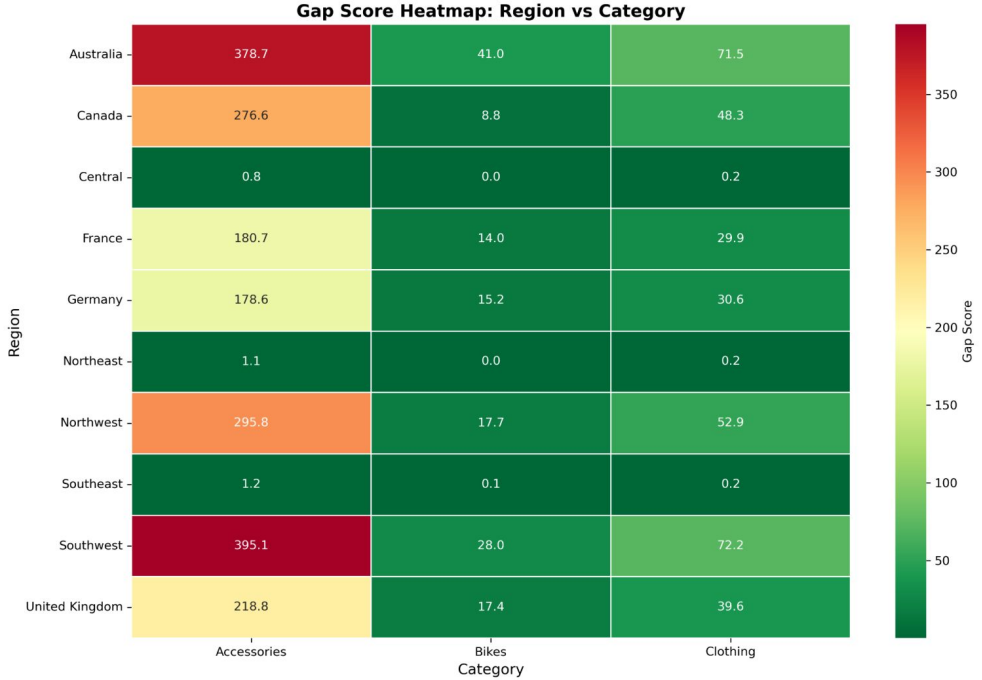
Tier	Score	Color	Action
Low Risk	0.0–0.49	Green text	Normal checkout
Medium	0.50–0.74	Yellow text	Extra verification
High Risk	0.75–1.0	Red text	Blocked, admin review

This is a behavioral safety layer — not the core of the platform. It supports human decision-making, not replaces it.

Gap Analysis

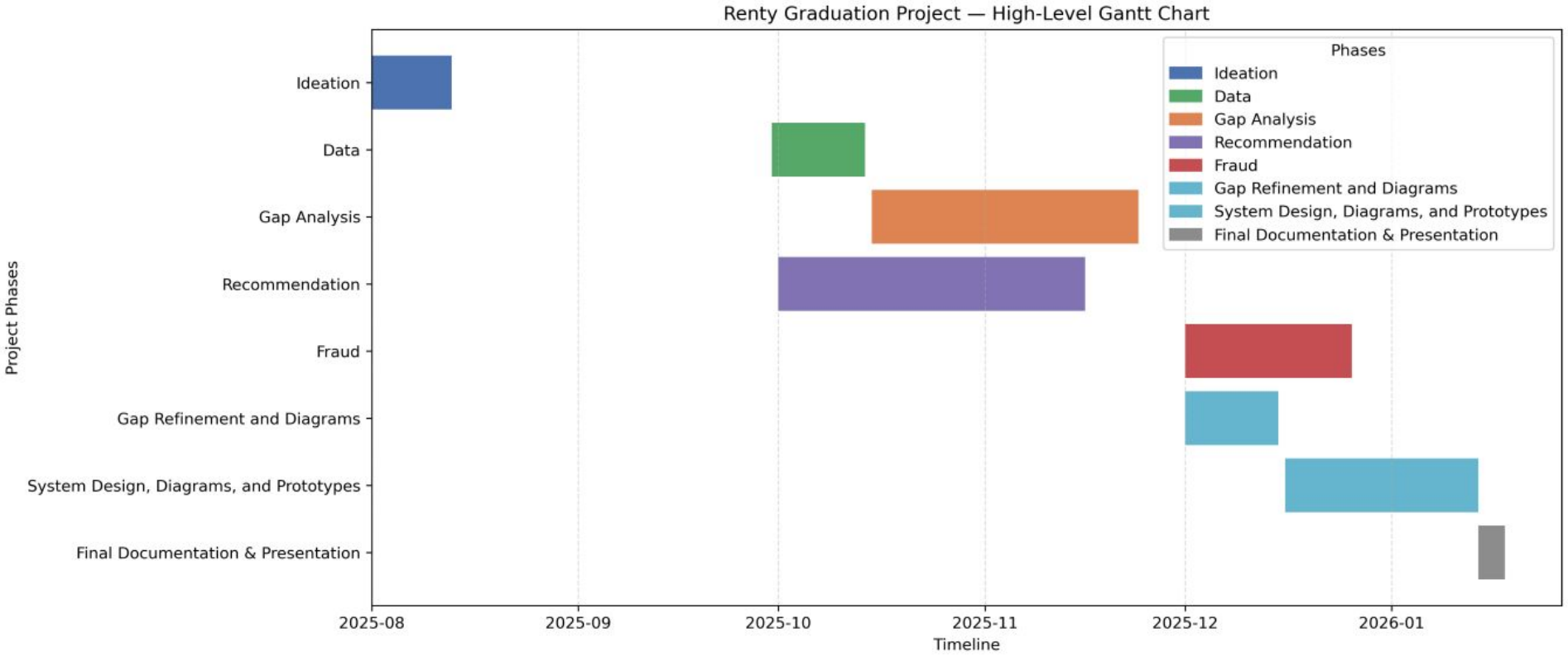
Gap Analysis

Gap Score	Severity	Action
< 50	***Low***	Monitor
50-100	***Moderate***	Incentivize owners
100-200	***High***	Active recruitment
> 200	***Critical***	Emergency campaign



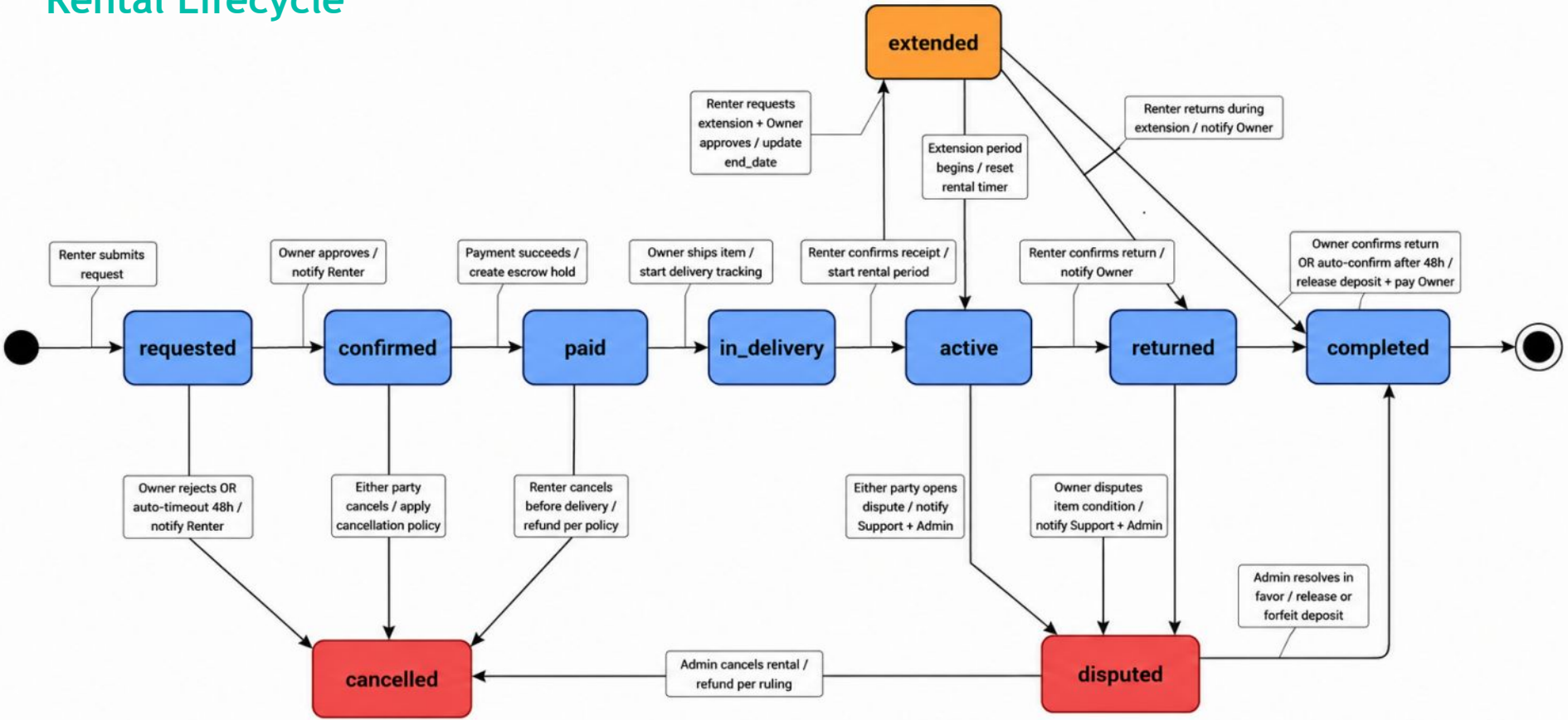
Semester 1 Timeline

Semester 1 Timeline



Rental Lifecycle

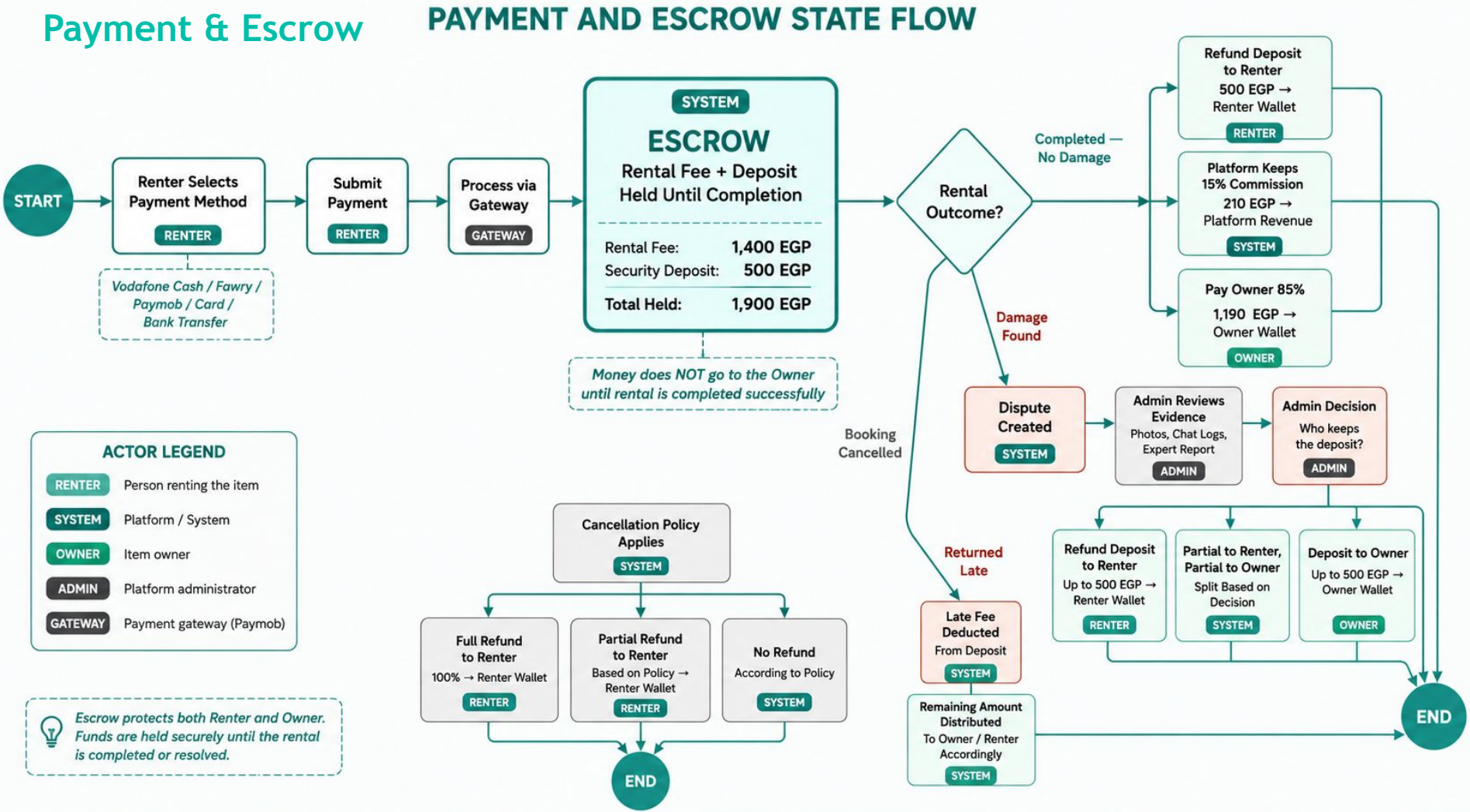
Rental Lifecycle



Every state transition is enforced at the database level using PostgreSQL ENUMs and check constraints.

Payment & Escrow

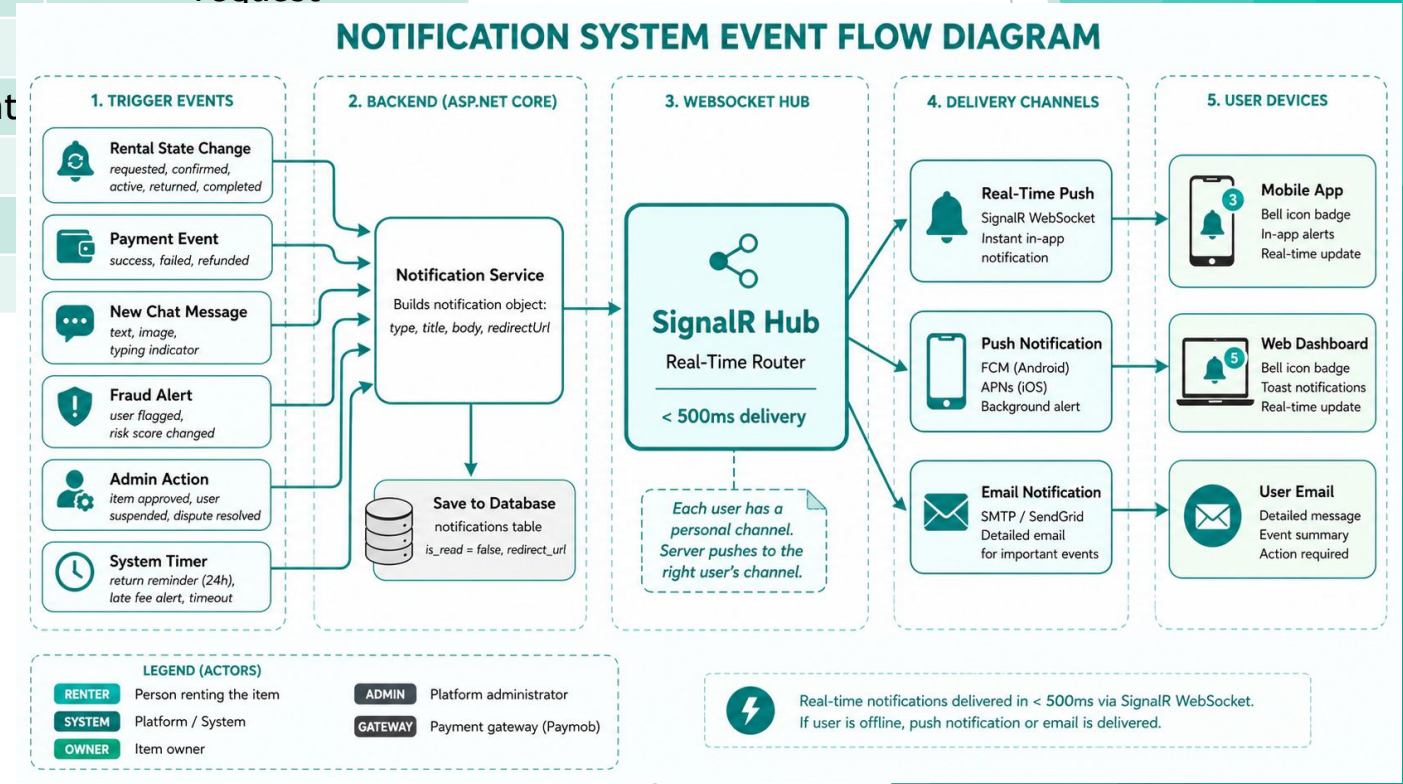
Payment & Escrow



Real Time Communication

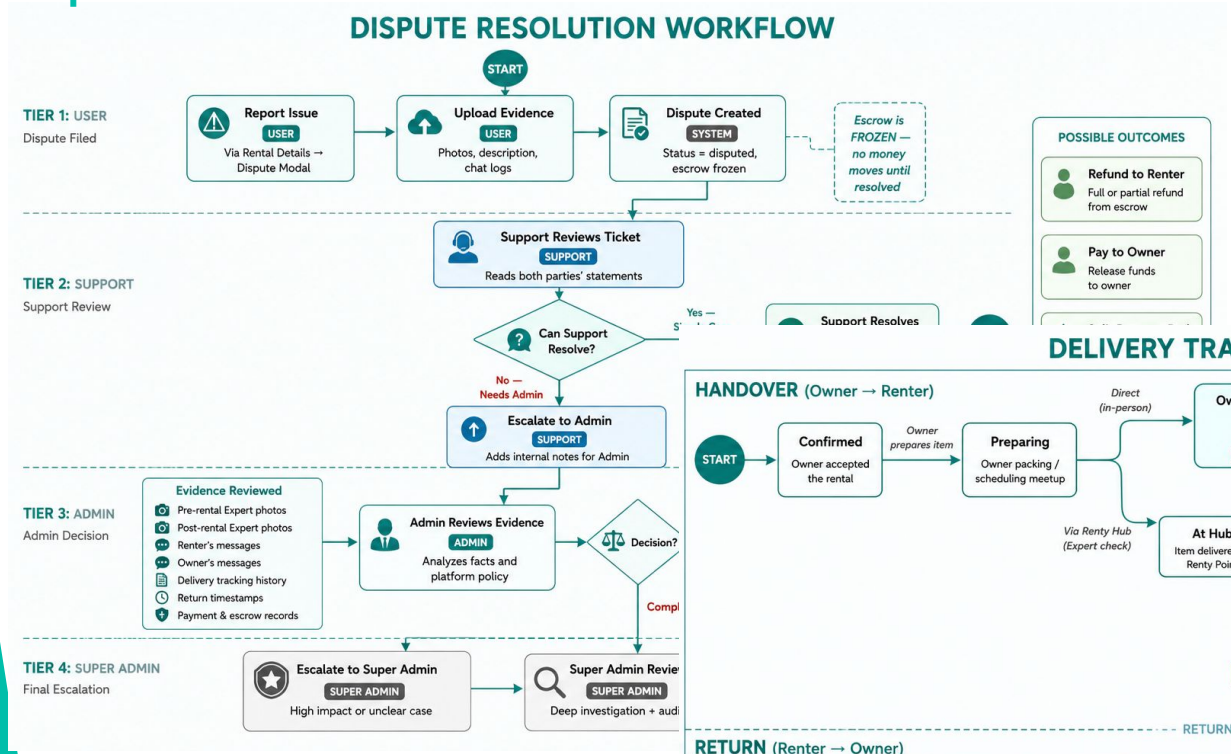
Real Time Communication

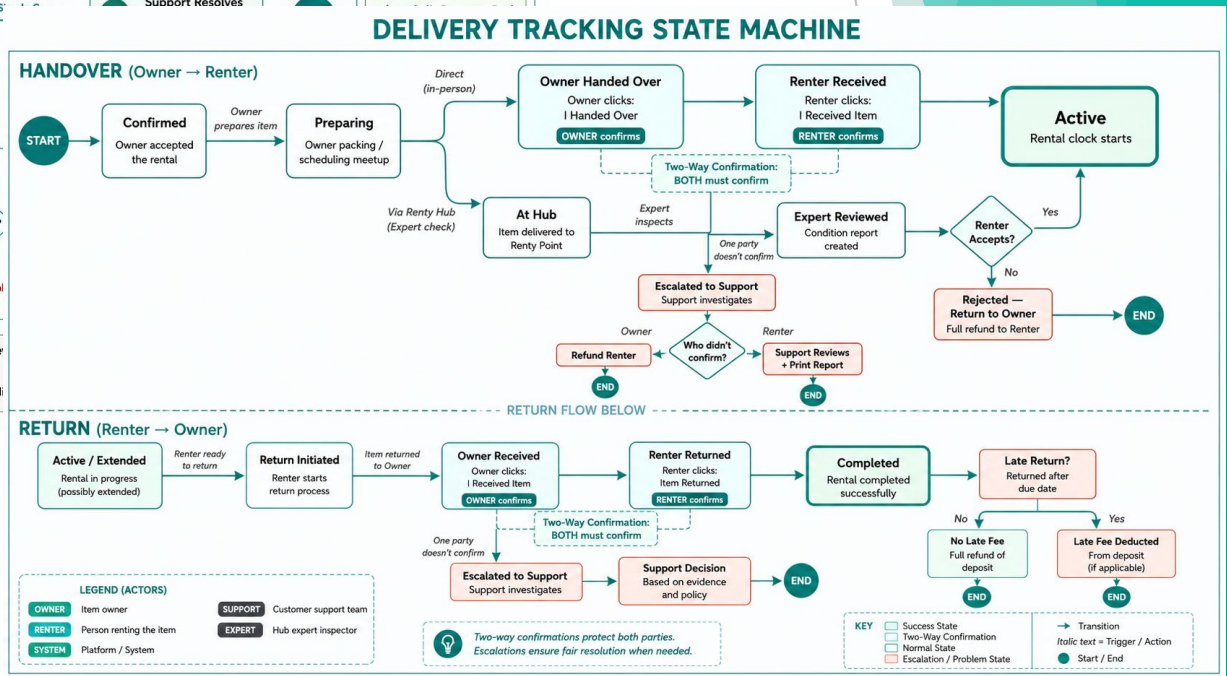
Event	Recipient	Trigger
Rental Confirmed	Renter	Owner approves request
Rental Started	Renter	
New Message	Recipient	
Return Reminder	Renter	
Late Fee Alert	Renter	
Dispute Opened	Both	



Operational Workflows

Operational Workflows





Trust Architecture

Trust Architecture

Escrow Payments

- All money held by platform until rental completes
- Renter protected: refund if owner doesn't deliver

Identity & Ratings

- Email + phone + ID verification
- Dual rating system (renter rates owner, owner rates renter)
- Fraud scoring as additional safety laye



Expert Verification

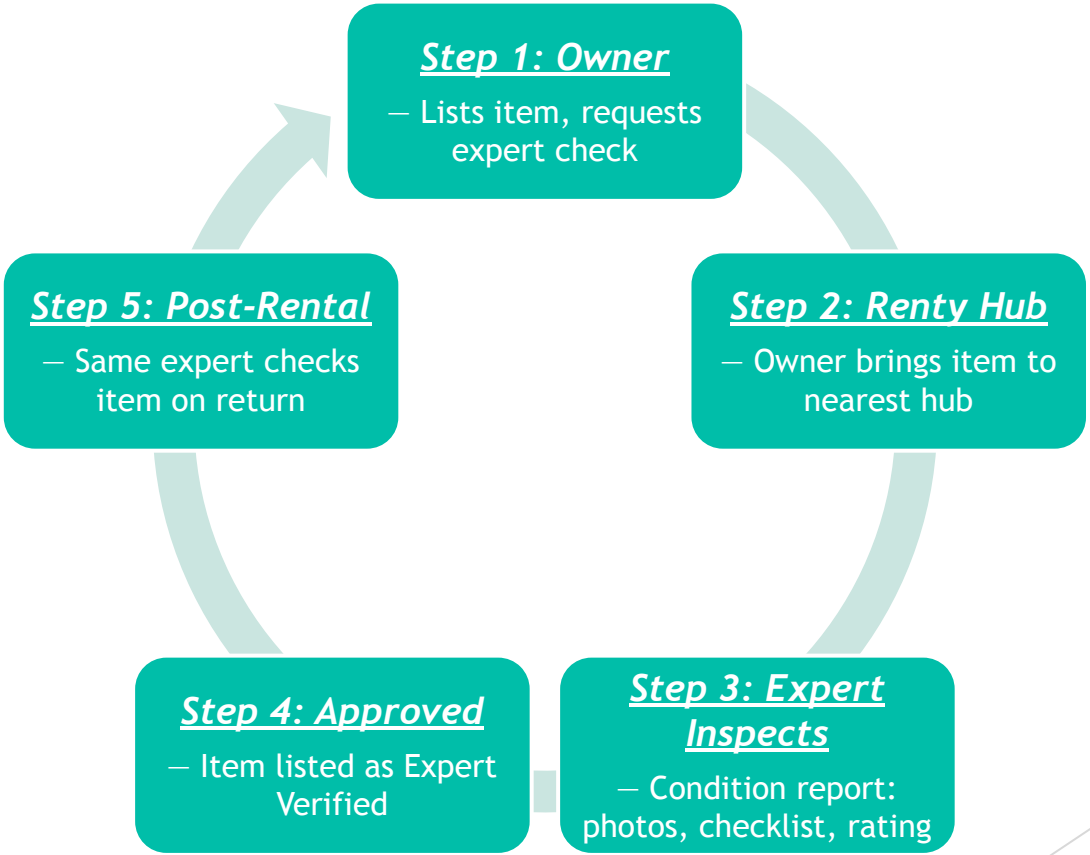
- Physical inspection at Renty Hubs
- Pre-rental and post-rental condition reports
- Same expert rule: consistency prevents

Dispute Resolution

- 3-tier escalation: Support > Admin > Super Admin
- Evidence-based: photos, chat logs, expert reports
- Cancellation policies: Flexible / Moderate / Strict

Expert Hubs

Expert Hubs



Continuity Rule: The SAME expert must perform both pre-rental and post-rental checks to prevent collusion and ensure accountability.

Smart Recommendations

Smart Recommendations

The Weekend Cyclist

- Friday 4PM: App opened
  — System detects weekend pattern
- Friday 5PM: Recommends Mountain Bike + Helmet bundle
- Saturday: User books
  — System logs interaction
- Next Friday: Proactive notification
  — Ready for this weekend?

Pattern: Every Fri-Sun --> target recommendations Friday afternoon

The Diving Tourist

- Tourist arrives in Dahab (detected via location/language)
- System recommends: Diving package
  — BCD + Regulator + Wetsuit
- Season: Summer
  — Water sports items promoted
- Language: English
  — Content served in English

Context-aware: Location + Season + Language = precise recommendations

Non-Functional Requirements

Non-Functional Requirements

Security

- JWT + Refresh Tokens (15min / 7day)
- RBAC: 6 roles, 7 policies
- FluentValidation on every request
- HTTPS/TLS 1.3 encryption
- Parameterized queries (SQL injection prevention)

Performance

- API response: < 200ms (p95)
- WebSocket delivery: < 500ms
- Angular lazy loading
- EF Core query optimization
- CDN for images (Cloudinary)

Scalability

- Clean Architecture (4 layers)
- CQRS pattern (read/write separation)
- Connection pooling
- Background services for batch jobs
- Modular monolith (microservice-ready)

Localization

Non-Functional Requirements

Language Strategy

- Primary: English (sports tourism — Dahab, Red Sea, Sinai)
- Secondary: Arabic (local Egyptian users)
- Architecture: Full i18n support built in
- RTL layout support ready

Currency Strategy

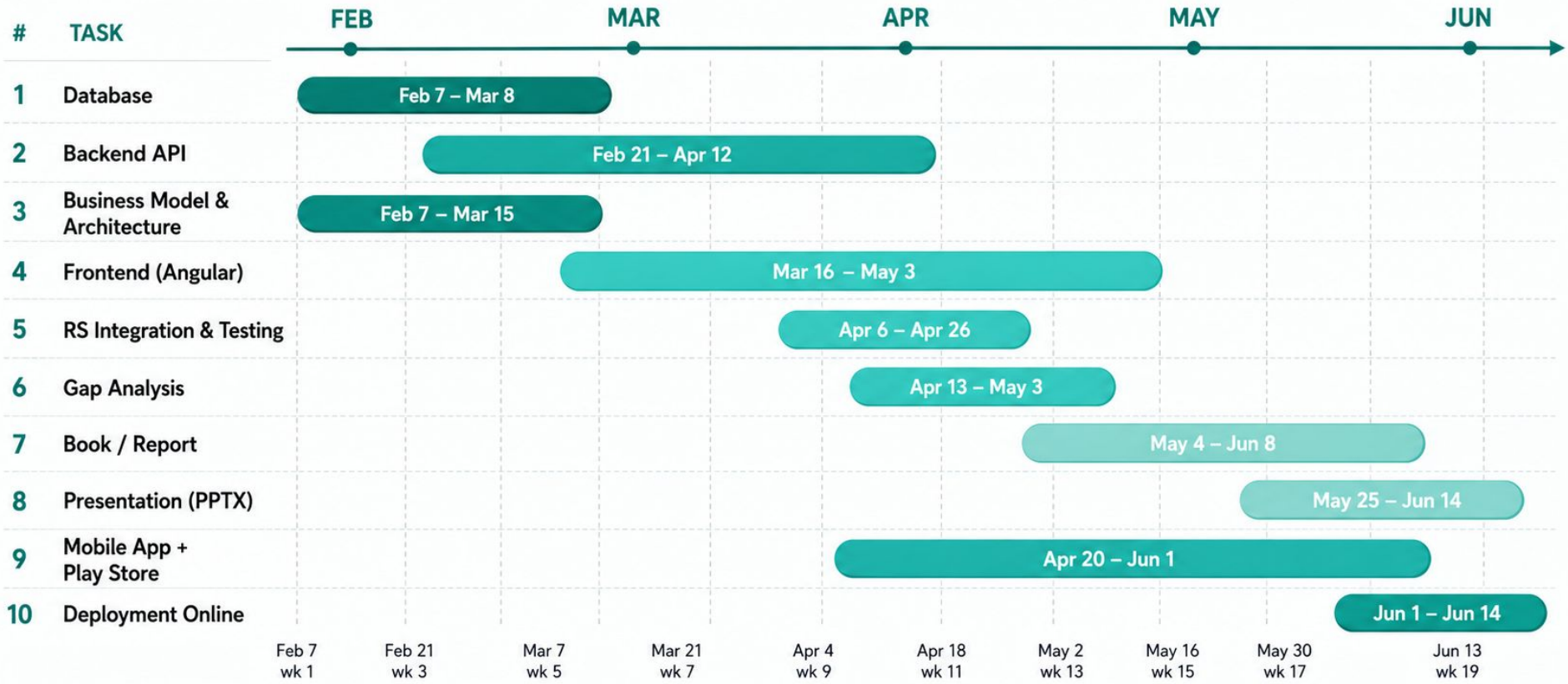
- Primary storage: EGP (Egyptian Pound)
- Tourist display: USD/EUR conversion at checkout
- Payment gateway: Paymob (international cards)
- Owner always paid in EGP

Example: Climbing rope 200 EGP/day --> Tourist sees ~$4 USD/day

Semester 2 Timeline

Semester 2 Timeline

Feb 7 – Jun 14, 2026



LEGEND: Development | Analysis & Integration | Documentation & Presentation | Release & Deployment

Tasks overlap to maximize parallel development and team productivity.

Thank you

Represented By :

- Yasser Ashraf Mohammed
- Hamza Hussain Youssef
- Ahmad Khlifa Abdulraouf
- Moaz Moustafa Abdul Hamid
- Moamen Ahmed Mohammed

Supervised By

- Dr Mohammed Khamis